AFEMA DEOGRACIOUS

Software Engineer

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LinkedIn • GitHub • Bitbucket • Stack Overflow • Personal Website

SUMMARY

Software engineering graduate and experienced practitioner holding an M.Sc. in Data Communication and Software Engineering from Makerere University with over 7 years of full-stack software development experience across distributed systems, database architectures, and agentic AI workflows.

EDUCATION

Master of Science in Data Communication and Software Engineering — Makerere University *Kampala • March 2021 – November 2025*

- Specialised in Software Engineering: Research topic “A model-driven serious game generator - A case of Android quiz game”. The study extended an existing game modelling language for use in “serious” game contexts through the identification and incorporation of new serious game features.

Bachelor's Degree of Information Technology — Mbarara University of Science and Technology *Mbarara • August 2014 – May 2017*

- Software development (Java, PHP), database design, and academic project work. Served as Student Leader.

Uganda Advanced Certificate of Education — St. Joseph’s College Ombaci *Arua • February 2012 – November 2013*

- Physics, Chemistry, and Biology.

Uganda Certificate of Education — St. Joseph’s College Ombaci *Arua • February 2008 – November 2011*

TECHNICAL DOMAINS

Distributed & Event-Driven Systems	Kafka, RabbitMQ, gRPC, Redis, Spring Microservices, Kubernetes, Docker
Databases	PostgreSQL, MySQL, TiDB, ClickHouse, MongoDB
Programming & Frameworks	Java/Spring Framework, PHP/Laravel, Angular, React / React Native, Node.js
Others	Requirements analysis, technical proposal & report writing, agentic workflows (Claude Code, Codex), project management, agile collaboration (Pivotal Tracker, Trello, Slack, Jira), Git version control

AWARDS & RESEARCH RECOGNITION

- Winner — Global Med-Tech Surgery Hackathon (CAMTECH Uganda, 2016): Recognized for innovative medical technology software solutions.
- Second Runner-Up Project of the Year — Mbarara University of Science and Technology (MUST): Awarded for exceptional undergraduate IT project research & implementation.

PROFESSIONAL & ENGINEERING EXPERIENCE

Software Engineer — pawaTech OU *Remote • May 2026 – Present*

- Integrating agentic coding tools (Claude Code and Codex) into modern software engineering workflows.
- Maintenance of Node.js backend infrastructure and architecture.
- Development of React front-end applications.
- Code reviews, testing, and quality control.
- Building CI/CD pipelines

Java Developer — betPawa *Kampala • April 2022 – May 2026*

- Maintenance and development of Spring-based low-latency real-time Java microservices handling high-throughput gaming event streams.
- System optimization and performance benchmarking across distributed event-driven systems (Kafka, RabbitMQ, Spring, Redis, MySQL, gRPC, Kubernetes, Docker).
- Unit testing using JUnit 5.
- Technical research.
- Systematic code reviews.
- Mentorship of junior colleagues.

Development Team Lead — NepServ Consults Limited *Kampala • March 2020 – March 2022*

- Technical team management and mentorship.
- Project management.
- System requirements analysis.
- Full-stack development using Java (Vanilla & Spring Boot), Angular, PostgreSQL, MySQL, PHP (Laravel), JavaScript, and React.
- Technical documentation, application server setup, and cloud deployment.

Software Developer — NepServ Consults Limited *Kampala • March 2019 – March 2022*

- Requirements analysis, technical specification writing, and multi-tier application development.

Associate, Operations Management — Imprint (U) Limited *Kampala • January 2018 – March 2019*

- Business proposal and technical report writing, facilitation of strategic planning, process documentation, records management, and IT support.

APPLIED SOFTWARE ENGINEERING PROJECTS

- **LOCKER ROOM BONUS:** A rewards system for local leagues in various African countries. I redeveloped the user interface from the non-responsive and non-intuitive initial version to a modern and responsive version and expanded features supported by the application.
- **REGULATORY REPORTING PLATFORM (7+ COUNTRIES):** Integrated core gaming management systems and regulatory reporting platforms to satisfy strict compliance and data reporting requirements across various African jurisdictions.
- **NEPSERV VANTAGE:** Designed an affordable platform for Microfinance Institutions featuring financial integrations with banking and mobile money (Java, Angular).
- **SEED PLATFORM:** Designed and developed an agricultural supply chain system for farmers incorporating digital wallet, bank, and mobile money payment gateways (Java, React, PHP).
- **SAGE SYSTEM:** Participated in the development of the national social benefits management and disbursement platform for the Ministry of Gender, Labour and Social Development (Java, Angular).
- **CHOICE MICROFINANCE LOANS BOT:** Implemented an automated conversational banking/loans engine via WhatsApp integrated with M-Pesa (Java 1.8).
- **PEEC ENERGY:** Built an IoT web portal monitoring security guard deployment and site metrics (Angular, Laravel, PostgreSQL).
- **P-MONITOR:** Created web and mobile platforms providing pregnancy tracking and healthcare informational delivery (PHP, Bootstrap).
- **PROFSEL:** Point of Sale software for SMEs built with Laravel (PHP) and Angular.
- **ABF CONTRIBUTION SCHEME PLATFORM:** Full-stack fundraising platform developed using Laravel and Angular.
- **SOKO UGANDA:** E-commerce platform supporting locally manufactured goods under the government Buy Uganda Build Uganda initiative (PHP, Bootstrap, MySQL, JavaScript).

STRATEGIC CONSULTING & NON-IT PROJECTS

- Development of business management tools for UMBA veterinary services under the Palladium NUTEC-MD program.
- Strategic planning facilitation for MARSH Uganda Limited and various organizational entities.
- Project management and technical consulting for international entities (Palladium, GIZ, Enabel / BTC, Vivo Energy).
- Establishment of an institutional knowledge management system (Imprint Uganda Limited).

VOLUNTEERING & COMMUNITY LEADERSHIP

- International Service Director — Rotaract Rubaga (2018–2019, 2020–2021): Facilitated international organizational partnerships and external visibility initiatives.
- Community Volunteer — Sanyu Babies Home, Kampala (2018).

LANGUAGES (CEFR STANDARD)

- English: Professional / Academic Mastery — Listening C1, Reading C1, Spoken Interaction C2, Spoken Production C2, Writing C1 (CEFR Framework).
- Lugbara: Native speaker (Proficient in speaking, intermediate in writing).
- Swahili & Luganda: Beginner level.

HOBBIES

- Jogging
- Cycling
- Reading local and international news.

REFERENCES

- Available upon request.